

CBSE Class 12 Mathematics — Chapterwise Formula Sheet

(Note: The chapter structure, syllabus topics, and exam guidelines align with the official CBSE Class 12 syllabus provided in the sources [1-10]. However, as the sources do not contain a complete mathematical formula sheet, the specific formulas, derivations, and standard identities provided below are drawn from my expert knowledge. You may want to independently verify them.)

Relations & Functions

Types of Relations: [3] For a relation R on set A : * **Reflexive:** $(a, a) \in R$ for every $a \in A$ * **Symmetric:** If $(a, b) \in R$, then $(b, a) \in R$ for all $a, b \in A$ * **Transitive:** If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$ for all $a, b, c \in A$ * **Equivalence Relation:** A relation that is reflexive, symmetric, and transitive.

Types of Functions: [3] * **One-One (Injective):** $f(x_1) = f(x_2) \implies x_1 = x_2$ for all $x_1, x_2 \in \text{Domain}$. * **Onto (Surjective):** For every $y \in \text{Codomain}$, there exists an $x \in \text{Domain}$ such that $f(x) = y$. * **Bijective Function:** A function that is both one-one and onto.

Inverse Trigonometric Functions

Domain and Principal Value Branches: [3] * $y = \sin^{-1} x \implies \text{Domain: } [-1, 1], \text{ Range: } [-\frac{\pi}{2}, \frac{\pi}{2}]$ * $y = \cos^{-1} x \implies \text{Domain: } [-1, 1], \text{ Range: } [0, \pi]$ * $y = \tan^{-1} x \implies \text{Domain: } \mathbb{R}, \text{ Range: } (-\frac{\pi}{2}, \frac{\pi}{2})$ * $y = \cot^{-1} x \implies \text{Domain: } \mathbb{R}, \text{ Range: } (0, \pi)$ * $y = \sec^{-1} x \implies \text{Domain: } \mathbb{R} - (-1, 1), \text{ Range: } [0, \pi] - \{\frac{\pi}{2}\}$ * $y = \csc^{-1} x \implies \text{Domain: } \mathbb{R} - (-1, 1), \text{ Range: } [-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$

Standard Properties:

$$\sin^{-1}(-x) = -\sin^{-1} x \quad ; \quad \cos^{-1}(-x) = \pi - \cos^{-1} x$$

$$\tan^{-1}(-x) = -\tan^{-1} x \quad ; \quad \cot^{-1}(-x) = \pi - \cot^{-1} x$$

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \quad ; \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$

Matrices

Types and Operations: [3] * **Equality:** $A = B \implies a_{ij} = b_{ij}$ for all i, j . * **Transpose (A^T)** **Properties:** $(A^T)^T = A$, $(kA)^T = kA^T$, $(A + B)^T = A^T + B^T$, $(AB)^T = B^T A^T$. * **Symmetric Matrix:** $A^T = A$ * **Skew-Symmetric Matrix:** $A^T = -A$ (Diagonal elements are

zero). * **Theorem:** Any square matrix can be expressed as the sum of a symmetric and skew-symmetric matrix:

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

Determinants

Formulas & Properties: [4] * **Area of Triangle:** With vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

* **Adjoint of Matrix ($adjA$):** Transpose of the cofactor matrix.

$$A \cdot (adjA) = (adjA) \cdot A = |A|I$$

* **Inverse of a Matrix:**

$$A^{-1} = \frac{1}{|A|} adj(A) \quad (\text{if } |A| \neq 0)$$

* **Solving Linear Equations (Matrix Method):** For $AX = B$, the unique solution is $X = A^{-1}B$.

Continuity & Differentiability

Continuity: [4] A function $f(x)$ is continuous at $x = a$ if:

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

Standard Derivatives: [5] * $\frac{d}{dx}(x^n) = nx^{n-1}$ * $\frac{d}{dx}(e^x) = e^x$; $\frac{d}{dx}(a^x) = a^x \ln a$ *
 $\frac{d}{dx}(\ln x) = \frac{1}{x}$ * $\frac{d}{dx}(\sin x) = \cos x$; $\frac{d}{dx}(\cos x) = -\sin x$ *
 $\frac{d}{dx}(\tan x) = \sec^2 x$; $\frac{d}{dx}(\cot x) = -\csc^2 x$ *
 $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$; $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

Rules of Differentiation: [11] * **Product Rule:** $(uv)' = u'v + uv'$ * **Quotient Rule:**

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2} \quad \text{* Chain Rule: } \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \quad \text{* Parametric Form: } \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Application of Derivatives

Key Concepts: [5] * **Rate of Change:** $\frac{dy}{dx}$ represents the rate of change of y w.r.t x . *

Increasing/Decreasing Functions: * Strictly Increasing if $f'(x) > 0$ * Strictly Decreasing if

$f'(x) < 0$ * **Maxima and Minima:** * *First Derivative Test:* Find critical points where $f'(x) = 0$. If $f'(x)$ changes from $+$ to $-$ across a point, it's a local maximum. If $-$ to $+$, it's a local minimum. * *Second Derivative Test:* If $f'(c) = 0$ and $f''(c) < 0$, c is a point of local maxima. If $f''(c) > 0$, c is a local minima.

Integrals

Standard Integrals Table: [6] * $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ ($n \neq -1$) * $\int \frac{1}{x} dx = \ln|x| + C$ * $\int e^x dx = e^x + C$ * $\int \sin x dx = -\cos x + C$; $\int \cos x dx = \sin x + C$ * $\int \tan x dx = \ln|\sec x| + C$; $\int \cot x dx = \ln|\sin x| + C$ * $\int \sec x dx = \ln|\sec x + \tan x| + C$ * $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$ * $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$ * $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln\left|\frac{x-a}{x+a}\right| + C$ * $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln\left|\frac{a+x}{a-x}\right| + C$ * $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + C$

Integration by Parts (ILATE Rule): [6]

$$\int uv dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$$

(Order of preference for u : Inverse, Logarithmic, Algebraic, Trigonometric, Exponential)

Properties of Definite Integrals: [12] * $\int_a^b f(x) dx = \int_a^b f(t) dt$ * $\int_a^b f(x) dx = -\int_b^a f(x) dx$ * $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$ * $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ * $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ (if $f(x)$ is even) or 0 (if $f(x)$ is odd).

Application of Integrals

Area under Simple Curves: [6] * Area bounded by curve $y = f(x)$, x -axis, and ordinates $x = a$ to $x = b$:

$$\text{Area} = \int_a^b |y| dx = \int_a^b |f(x)| dx$$

* Area bounded by curve $x = g(y)$, y -axis, and lines $y = c$ to $y = d$:

$$\text{Area} = \int_c^d |x| dy = \int_c^d |g(y)| dy$$

Differential Equations

Formulas & Methods: [7] * **Order:** The order of the highest derivative. * **Degree:** The highest power of the highest order derivative (when written as a polynomial in derivatives). *

Variable Separable: $f(y)dy = g(x)dx \implies \int f(y)dy = \int g(x)dx + C$ * **Homogeneous**

DE: Substitute $y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$ * **Linear Differential Equation:** Form:

$\frac{dy}{dx} + Py = Q$ (where P, Q are functions of x) Integrating Factor (I.F.):

$$I.F. = e^{\int P dx}$$

General Solution:

$$y \cdot (I.F.) = \int (Q \cdot I.F.) dx + C$$

Vector Algebra

Standard Identities: [8] Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$. * **Magnitude:**

$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$ * **Unit Vector:** $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$ * **Dot (Scalar) Product:**

$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta = a_1b_1 + a_2b_2 + a_3b_3$ * **Projection:** Projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$ * **Cross**

(Vector) Product: $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin \theta \hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$ * **Area of Parallelogram:** Adjacent

sides $\vec{a}$ and $\vec{b} \implies \text{Area} = |\vec{a} \times \vec{b}|$

3D Geometry

Lines and Shortest Distance: [9] * **Direction Cosines:** $l^2 + m^2 + n^2 = 1$ [13] * **Equation of**

a Line (Vector Form): $\vec{r} = \vec{a} + \lambda \vec{b}$ * **Equation of a Line (Cartesian Form):**

$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$ * **Angle between Two Lines:** $\cos \theta = \left| \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1||\vec{b}_2|} \right|$ * **Shortest Distance**

Between Skew Lines: [14, 15] For lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$:

$$d = \left| \frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

* **Shortest Distance Between Parallel Lines:** For lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}$:

$$d = \left| \frac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|} \right|$$

Linear Programming

Core Concepts: [9] * **Objective Function:** A linear function $Z = ax + by$ which is to be maximized or minimized. * **Constraints:** Linear inequalities governing the resources. *

Corner Point Theorem: The optimal value of the objective function always occurs at one of the corner points of the bounded feasible region.

Probability

Key Formulas & Theorems: [10] * **Conditional Probability:** [16]

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (\text{if } P(B) \neq 0)$$

* **Multiplication Theorem:**

$$P(A \cap B) = P(A) \cdot P(B|A)$$

* **Independent Events:** [17]

$$P(A \cap B) = P(A) \cdot P(B)$$

* **Theorem of Total Probability:**

$$P(A) = \sum_{i=1}^n P(E_i)P(A|E_i)$$

* **Bayes' Theorem:**

$$P(E_i|A) = \frac{P(E_i) \cdot P(A|E_i)}{\sum_{j=1}^n P(E_j) \cdot P(A|E_j)}$$

* **Mean (Expected Value) of a Random Variable:**

$$\mu = E(X) = \sum_{i=1}^n x_i p_i$$

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